Mechanism, Thermal, and Structural Analysis with Validation

Watt I linkage path generation, heat-sink conduction–convection FEA, and stepped-shaft torsion FEA, each checked against an analytical model

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Course	ME 403 Engineering Design, NJIT	**Software**	Creo Parametric (mechanism, thermal), SolidWorks Simulation, MATLAB, Excel
Team	Samuel Cohen, Luis Ramirez, Larimel Rosario, Diego Arias, Miguel Huayahuas	**Inputs**	Link lengths, h, temperatures, and torque from the student ID (…87, …287)

PROJECT SUMMARY

Three CAD-based analyses were each validated against a closed-form model. A six-link Watt I mechanism was built in Creo and the paths of points p1 and q1 were traced over a full crank rotation; its four-bar sub-loop matched a MATLAB displacement model. A finned aluminum heat sink and a single 0.035 in fin were analyzed for steady conduction and convection, and the fin surface temperatures agreed with the analytical fin equation within about 11 °F (5%), falling from 228.7 °F at the base to about 97 °F at the tip. Finally, a stepped steel shaft under 128.7 in-lbf of torque was analyzed in SolidWorks; the peak fillet shear stress of 866 psi was within 5% of the stress-concentration hand calculation.

1. Introduction

Simulation results are only useful if they can be trusted. A standard practice is to validate a CAD or finite element model on a simpler case that has a known analytical answer, and only then use the model for the more complex geometry. This assignment applied that approach to three problems: a multi-loop linkage, a heat sink, and a shaft with a stress concentration.

2. Watt I Mechanism Path Generation

The Watt I mechanism is a six-link, single-degree-of-freedom planar linkage made of a four-bar loop (crank, ternary coupler, ternary follower, and ground) with a second loop formed by two ternary intermediate links. The crank is 1.0, the coupler, follower, and ground are 1.25, and the two intermediate links have sides of 1.687 and 1.987. The model was assembled in Creo with revolute joints and a cylinder joint at the follower pivot and driven through one full crank revolution to trace points p1 and q1.

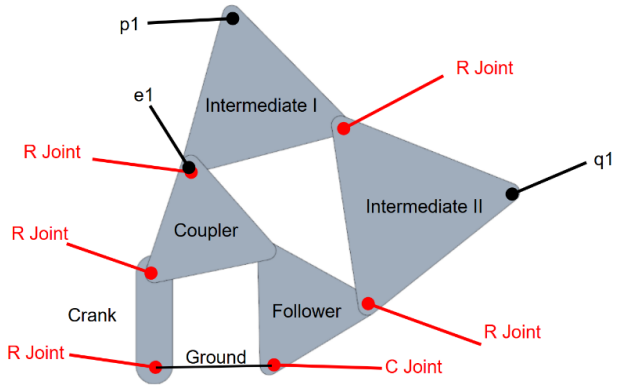


Figure 1. Creo model of the Watt I mechanism

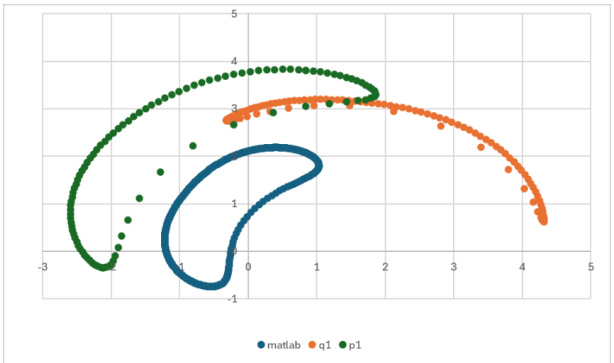


Figure 2. Paths of p1, q1, and e1 over one crank revolution

To validate the model, the four-bar loop was isolated and its coupler point e1 was traced in Creo and with the analytical complex-number displacement model in MATLAB, using crank W1 = 1 at 90°, coupler V1 = 1.25 at 11.5251°, follower U1 = 1.25 at 91.1553°, ground G1 = 1.25, and coupler point L1 = 1.25 at 71.5251°, stepping the crank 1° at a time.

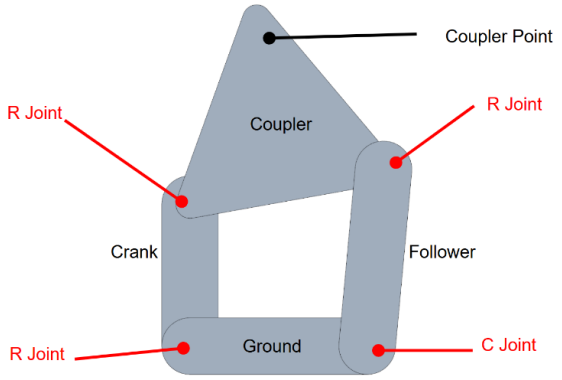


Figure 3. Creo model of the four-bar validation loop

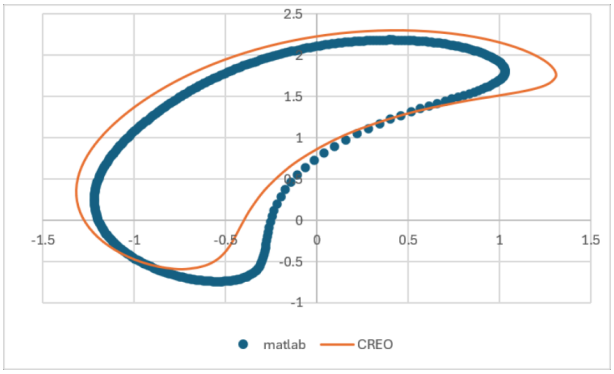


Figure 4. Coupler curve of e1: MATLAB vs. Creo

The two coupler curves share the same kidney shape, orientation, and extent, which confirms that the link lengths, joint definitions, and starting position in the Creo model are correct. The small offset between the curves comes from where the coupler point was picked on the Creo body relative to the ideal pin-center geometry used in MATLAB.

3. Heat Sink Thermal Analysis

An aluminum heat sink with a 1.5 in × 1.5 in thermal load applied under its base was modeled in Creo for steady conduction with convection from every exposed surface. As a validation case, a single rectangular fin 0.035 in thick, 1.95 in deep, and 1.30 in long was modeled with its base held at T_b = 228.7 °F, a convection coefficient of h = 0.0001287 BTU/s·in²·°F, and ambient air at T_∞ = 62.87 °F. Its surface temperatures were compared with the fin equation for convection at the tip:

$T(z) = T_\infty + (T_b - T_\infty)\cdot[\cosh m_1(L - z) + m_2 \sinh m_1(L - z)] / [\cosh m_1 L + m_2 \sinh m_1 L]$, with $m_1 = \sqrt{(2h(\text{depth} + t)/kA)} = 1.707$ /in and $m_2 = h/(m_1 k) = 0.0293$, using k = 9266 lbm·in/s³·°F (aluminum, about 192 W/m·K).

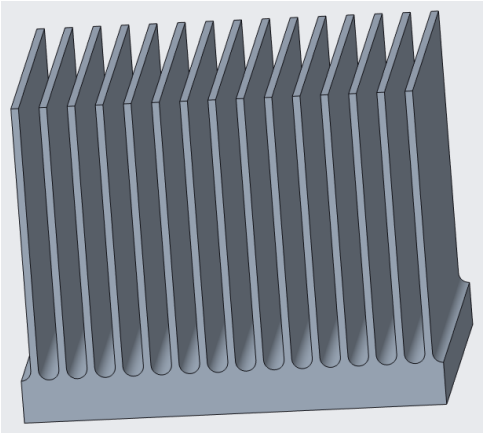


Figure 5. Creo model of the finned aluminum heat sink.

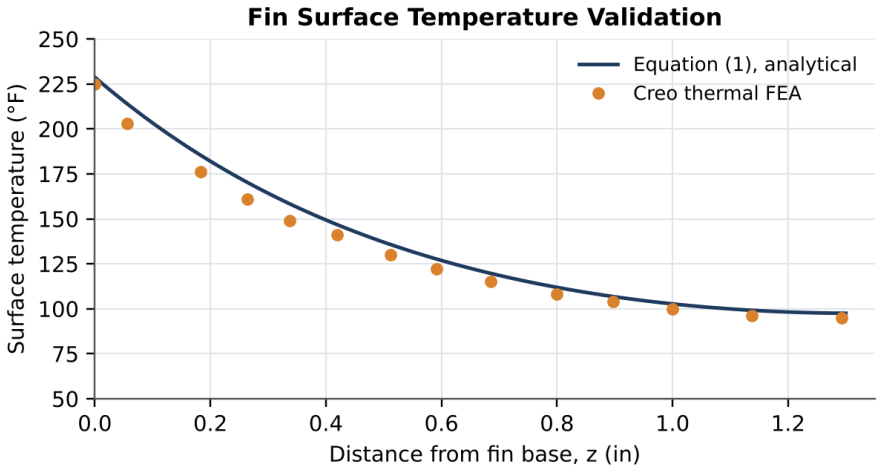


Figure 6. Fin surface temperature from the analytical equation and the Creo thermal FEA.

Table 1. Analytical fin surface temperature

z (in)	0.0	0.2	0.4	0.6	0.8	1.0	1.2
T (°F)	228.7	182.0	149.4	126.9	111.9	102.7	98.2

The FEA and analytical curves agree closely, with the FEA running 3–11 °F (at most about 5%) below the equation. The largest gap is in the first 0.3 in of the fin, where the 3D model lets heat spread through the base and the equation assumes purely one-dimensional conduction; toward the tip the two agree within about 3 °F. With the fin model validated, the same material, convection, and mesh settings were applied to the full heat sink.

4. Shaft Torsion with a Stress Concentration

A steel shaft stepping from a 1.2 in to a 1.0 in diameter with a 0.1 in fillet was fixed at one end and loaded with a torque of 128.7 in-lbf about its axis at the other. The analytical peak shear stress is $\tau = K_{ts} Tr/J$. With D/d = 1.2 and r/d = 0.1, the torsional stress-concentration factor is about $K_{ts} = 1.26$, and the nominal stress in the small section is $16T/\pi d^3 = 16(128.7)/\pi(1.0)^3 = 655.5$ psi, giving $\tau_{max} \approx 1.26 \times 655.5 = 826$ psi.

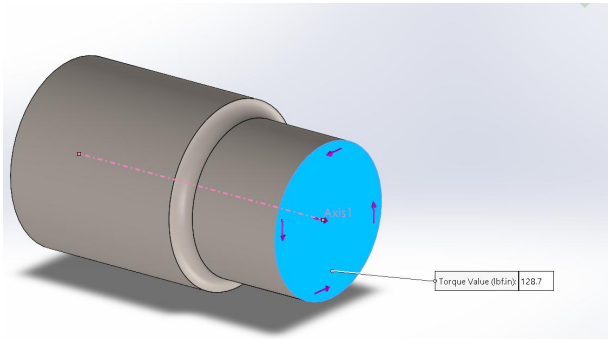


Figure 7. Shaft geometry, fixed end, and applied torque

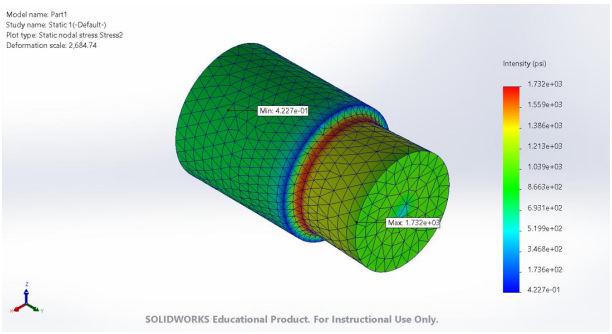


Figure 8. SolidWorks stress-intensity plot

Table 2. Shaft torsion results

Quantity	Value
Nominal shear stress, $16T/\pi d^3$	655.5 psi
Analytical peak shear stress, $K_{ts}Tr/J$	826 psi
FEA peak stress intensity ($\sigma_1 - \sigma_3$)	1,732 psi
FEA peak shear stress, $(\sigma_1 - \sigma_3)/2$	866 psi
Difference, FEA vs. analytical	4.8%

The SolidWorks plot shows the highest stress concentrated in a ring at the fillet, exactly where the stress concentration factor predicts it. Because stress intensity equals twice the maximum shear stress, the FEA peak of 1,732 psi corresponds to τ_{max} = 866 psi, 4.8% above the hand calculation. The remaining difference comes from reading K_{ts} off a chart and from mesh size at the fillet, which a local mesh refinement would reduce.

5. Conclusion

All three CAD models were validated against analytical results: the Watt I four-bar loop reproduced the MATLAB coupler curve, the single-fin thermal model matched the fin equation within about 5%, and the shaft FEA predicted the fillet shear stress within 4.8% of the stress-concentration calculation. These checks give confidence in the Watt I path curves and the full heat-sink temperature distribution, which have no simple closed-form solution.

References

Bergman, T. L., Lavine, A. S., Incropera, F. P., & DeWitt, D. P. (2017). *Fundamentals of heat and mass transfer* (8th ed.). Wiley.

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